



成都信息工程大学
Chengdu University of Information Technology

计算机图形学-第8章 三维用户界面

Computer Graphics



第八章 三维用户界面

- 
- 一、Canvas
 - 二、Basic Layout
 - 三、Interaction Components
 - 四、Animation Integration
 - 五、Multiple Resolutions
 - 六、Creating a World Space UI

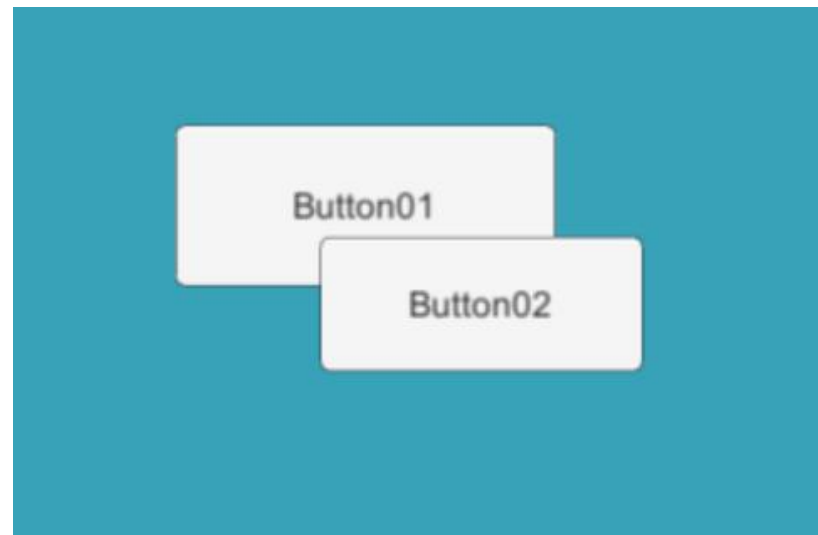
一、Canvas（画布）

- n** The **Canvas** is the area that all UI elements should be inside. The Canvas is a Game Object with a Canvas component on it, and all UI elements must be children of such a Canvas.
- n** **Canvas** uses the EventSystem object to help the Messaging System.

一、Canvas（画布）

n Draw order of elements

- n UI elements in the Canvas are drawn in the same order they appear in the Hierarchy. The first child is drawn first, the second child next, and so on. If two UI elements overlap, the later one will appear on top of the earlier one.



一、Canvas（画布）

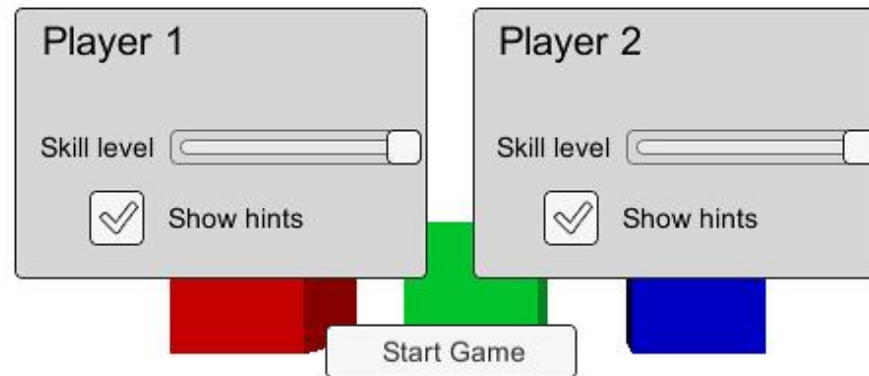
n **Render Modes**

- n The Canvas has a **Render Mode** setting which can be used to make it render in screen space or world space.

一、Canvas（画布）

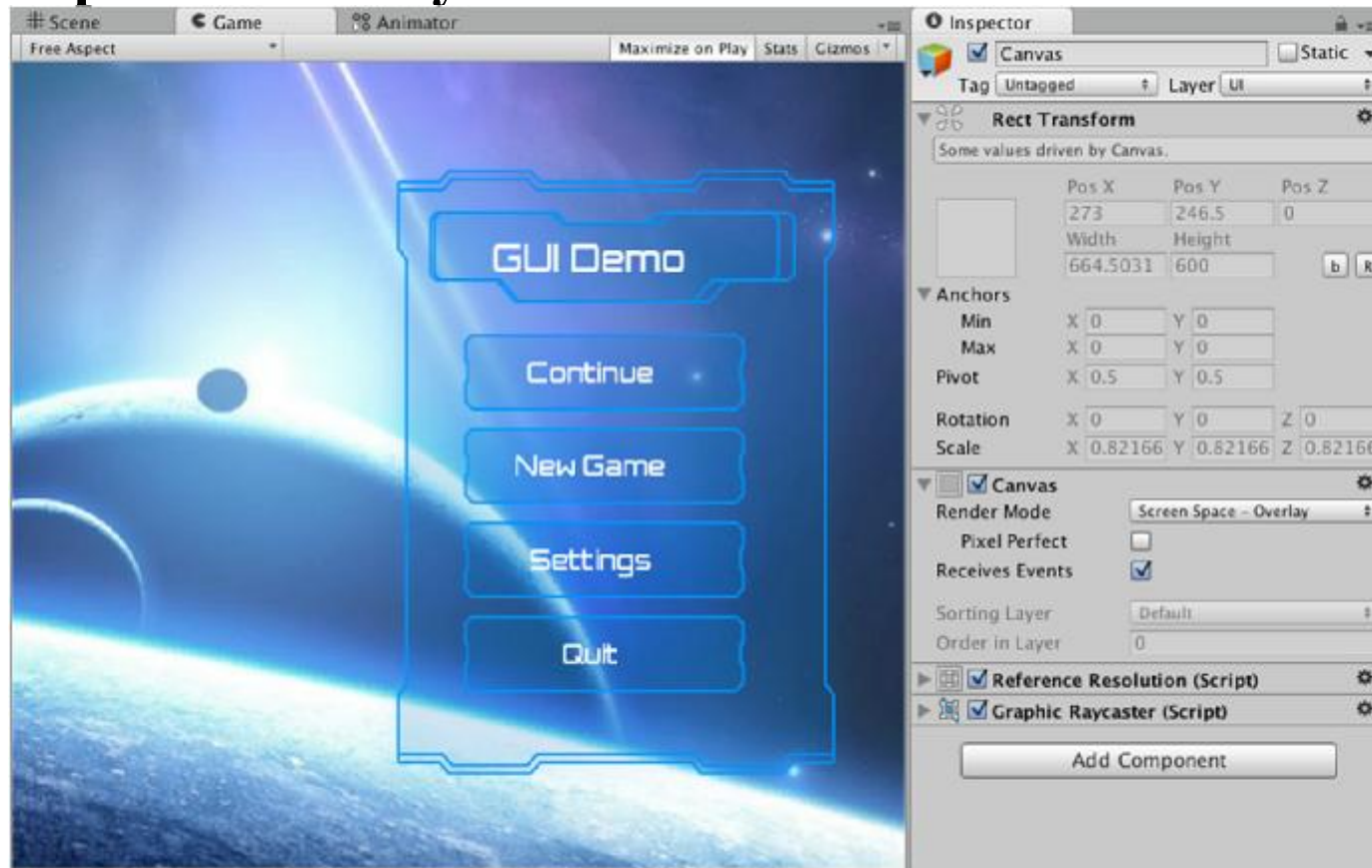
① Screen Space - Overlay

- n This render mode places UI elements on the screen rendered on top of the scene. If the screen is resized or changes resolution, the Canvas will automatically change size to match this.



一、Canvas (画布)

① Screen Space - Overlay



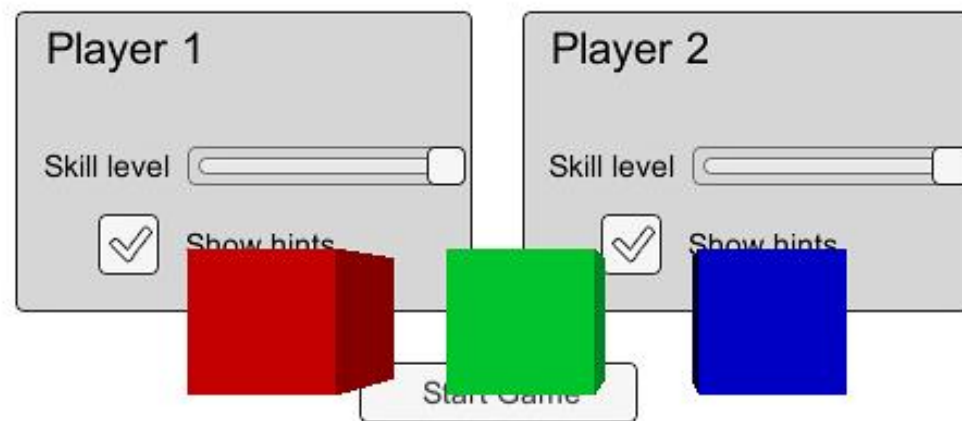
一、Canvas（画布）

② Screen Space - Camera

n This is similar to **Screen Space - Overlay**, but in this render mode the Canvas is placed a given distance in front of a specified **Camera**. The UI elements are rendered by this camera, which means that the Camera settings affect the appearance of the UI. If the Camera is set to **Perspective**, the UI elements will be rendered with perspective, and the amount of perspective distortion can be controlled by the Camera **Field of View**. If the screen is resized, changes resolution, or the camera frustum changes, the Canvas will automatically change size to match as well.

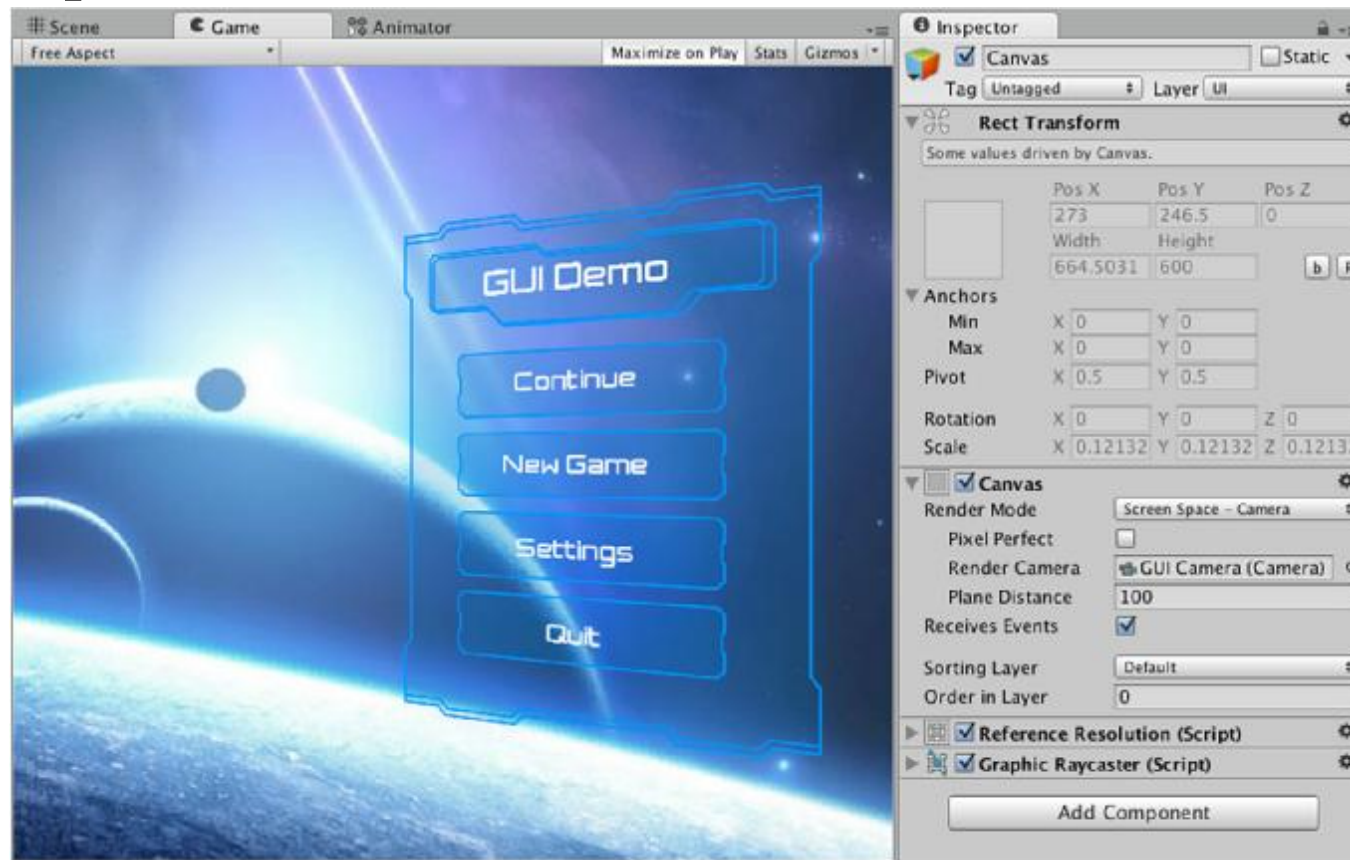
一、Canvas（画布）

② Screen Space - Camera



一、Canvas (画布)

② Screen Space - Camera



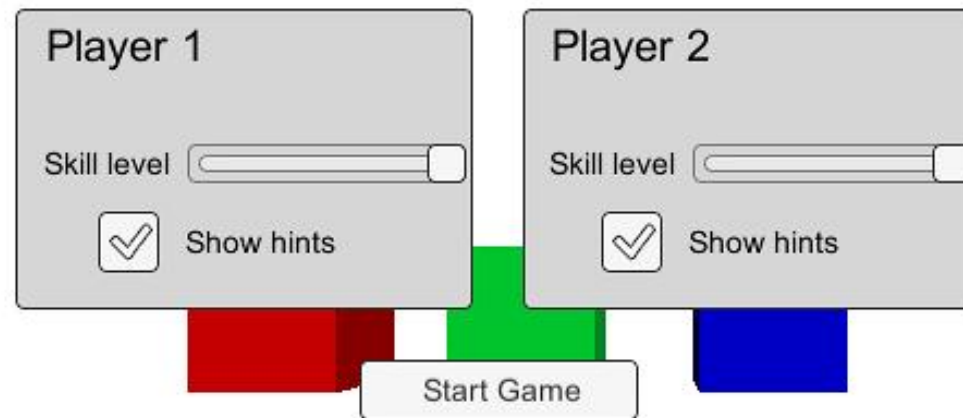
一、Canvas（画布）

③ World Space

- n In this render mode, the Canvas will behave as any other object in the scene. The size of the Canvas can be set manually using its Rect Transform, and UI elements will render in front of or behind other objects in the scene based on 3D placement. This is useful for UIs that are meant to be a part of the world. This is also known as a “diegetic interface”.

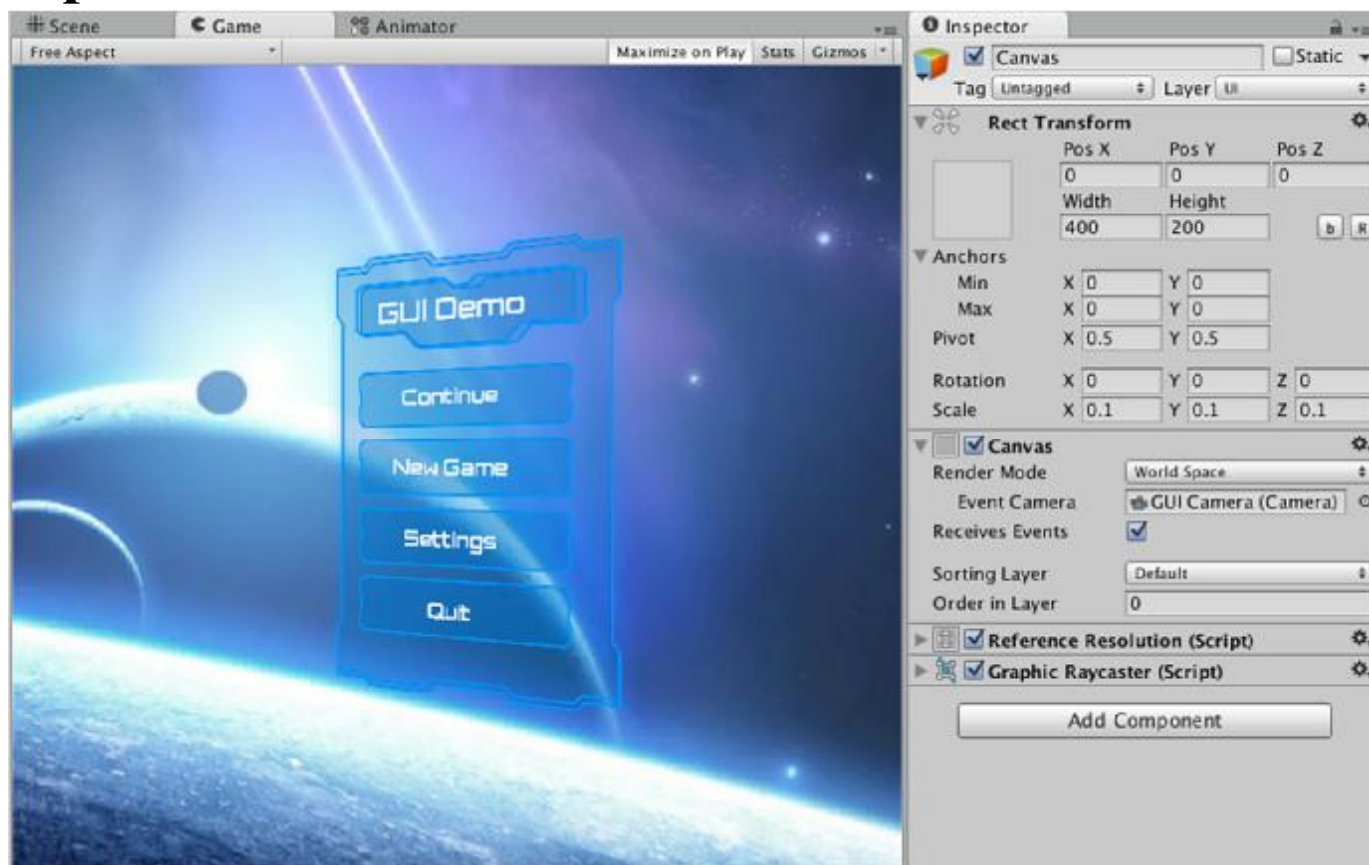
一、Canvas（画布）

③ World Space



一、Canvas (画布)

③ World Space



二、 Basic Layout

① The Rect Tool

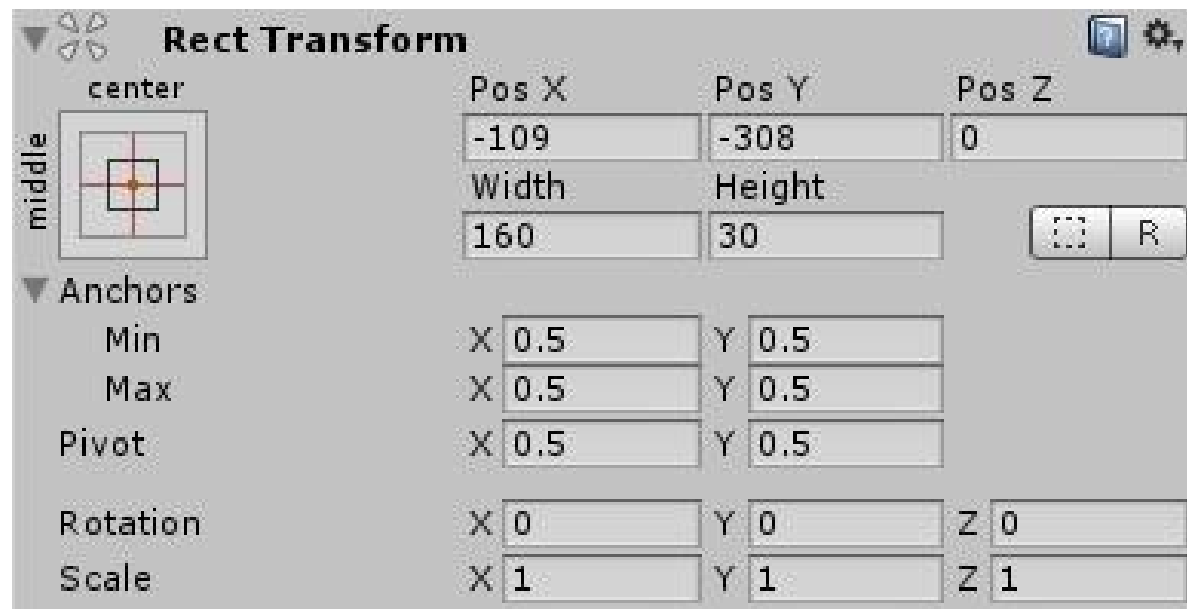
- n Every UI element is represented as a rectangle for layout purposes. This rectangle can be manipulated in the Scene View using the Rect Tool in the toolbar. The Rect Tool can be used to move, resize and rotate UI elements.



二、Basic Layout

② Rect Transform

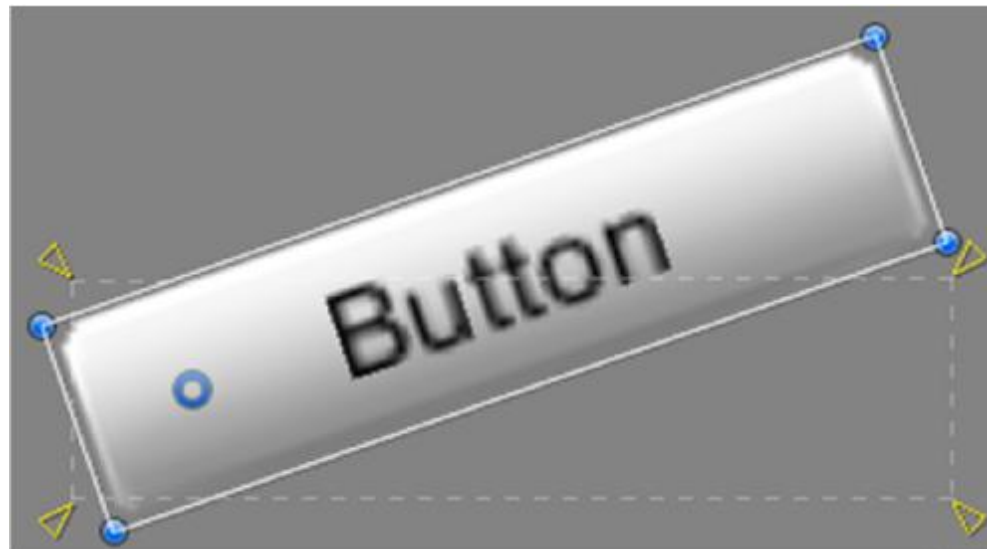
- n The Rect Transform is a new transform component that is used for all UI elements instead of the regular Transform component.



二、Basic Layout

③ Anchors

- n Rect Transforms include a layout concept called anchors. Anchors are shown as four small triangular handles in the Scene View.



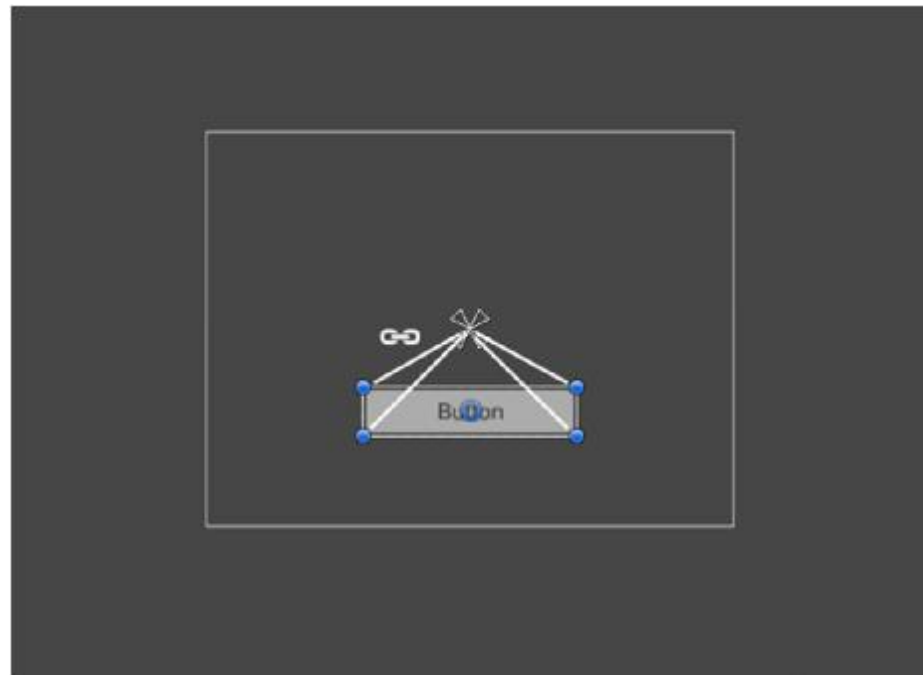
二、Basic Layout

③ Anchors

- n If the parent of a Rect Transform is also a Rect Transform, the child Rect Transform can be anchored to the parent Rect Transform in various ways. For example, the child can be anchored to the center of the parent, or to one of the corners.

二、Basic Layout

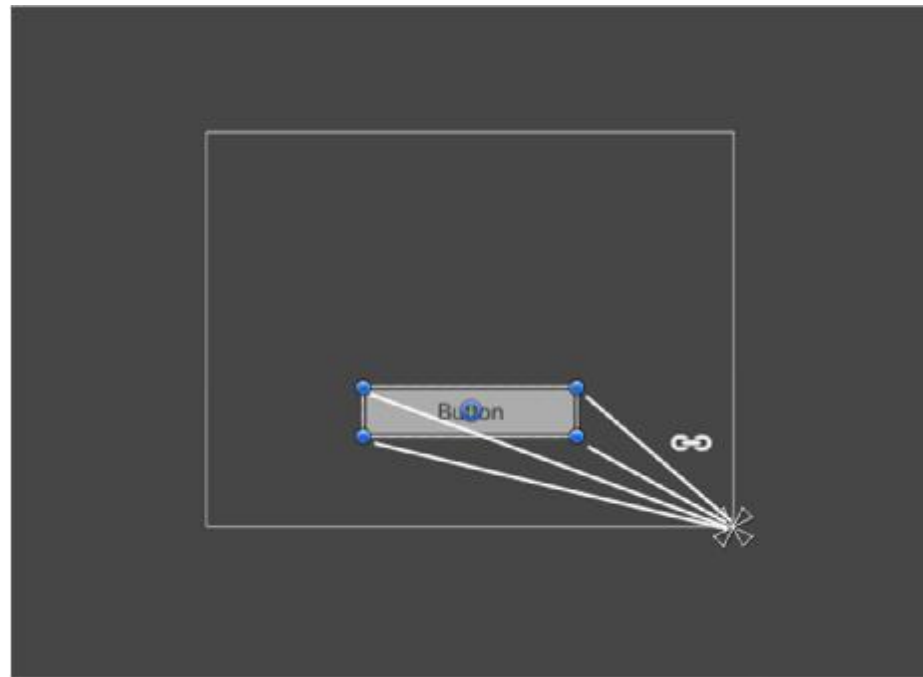
③ Anchors



UI element anchored to the center of the parent. The element maintains a fixed offset to the center.

二、Basic Layout

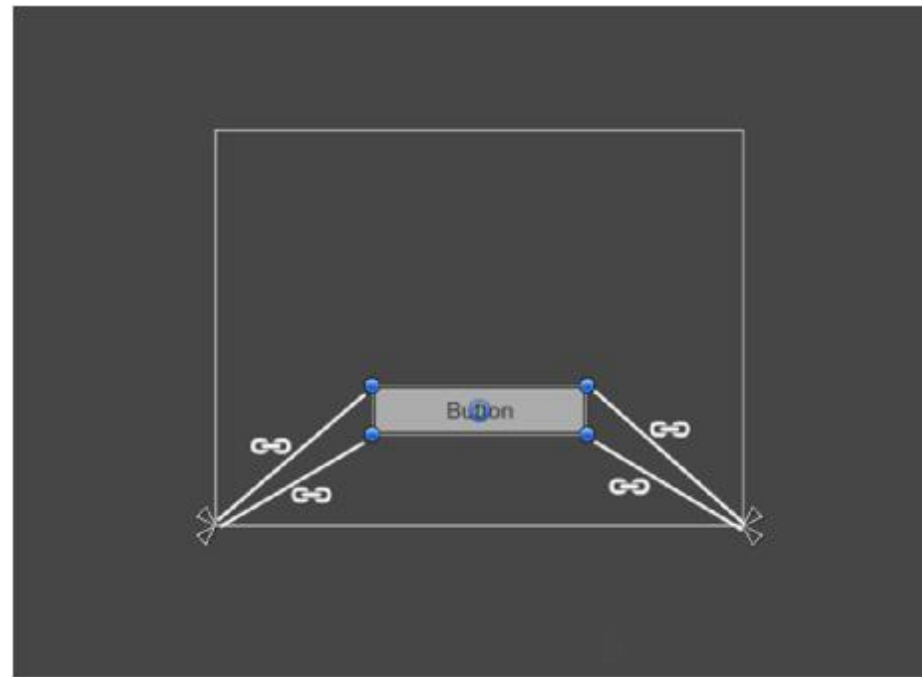
③ Anchors



UI element anchored to the lower right corner of the parent. The element maintains a fixed offset to the lower right corner.

二、Basic Layout

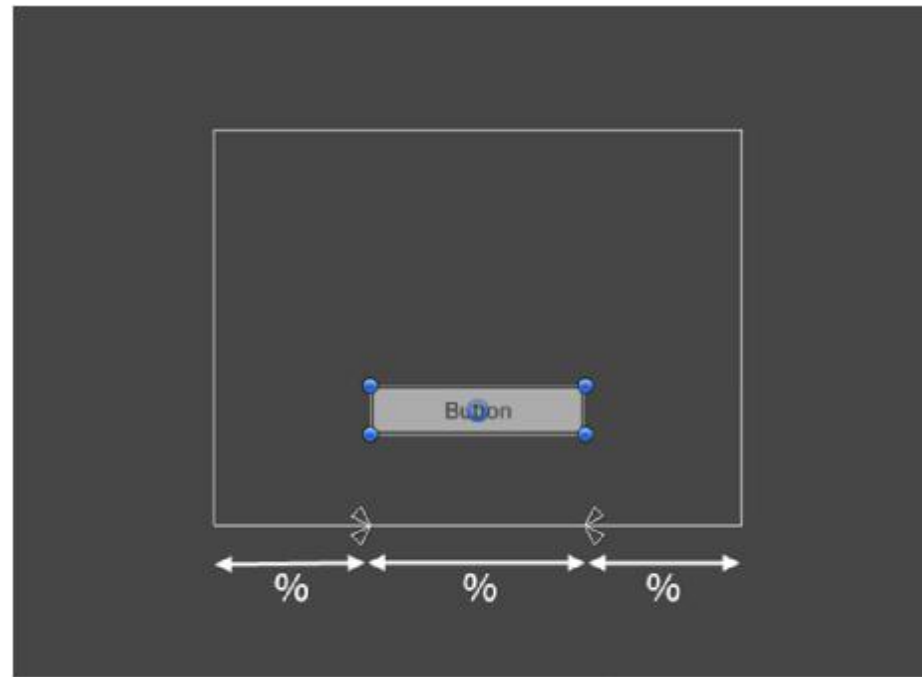
③ Anchors



UI element with left corners anchored to lower left corner of the parent and right corners anchored to lower right. The corners of the element maintains fixed offsets to their respective anchors.

二、Basic Layout

③ Anchors

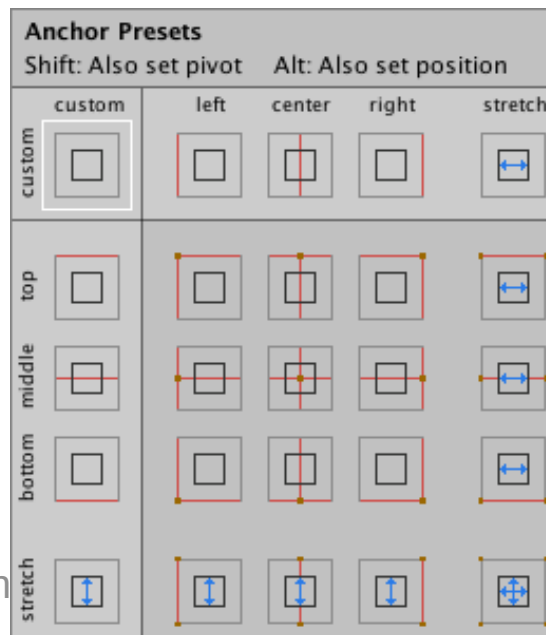


UI element with left corners anchored to a point a certain percentage from the left side of the parent and right corners anchored to a point a certain percentage from the right side of the parent rectangle.

二、Basic Layout

④ Anchor presets

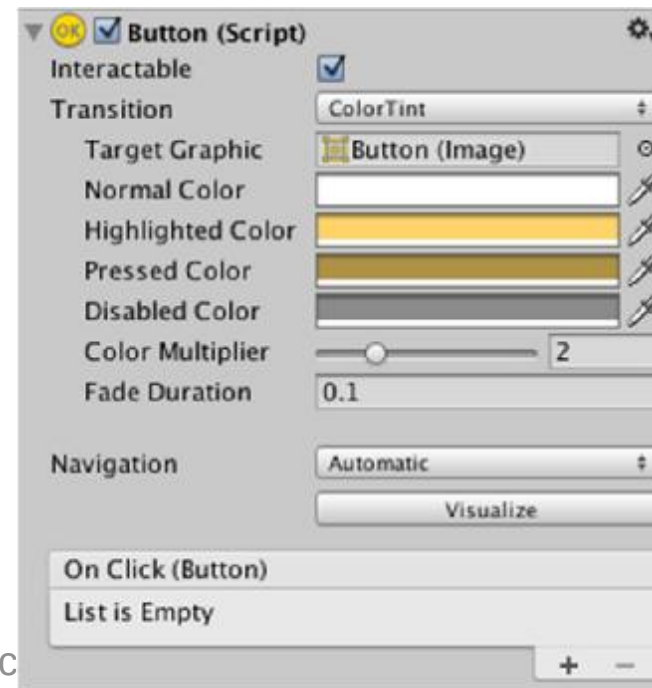
- n You can anchor the UI element to the sides or middle of the parent, or stretch together with the parent size. The horizontal and vertical anchoring is independent.



三、Interaction Components

① Button

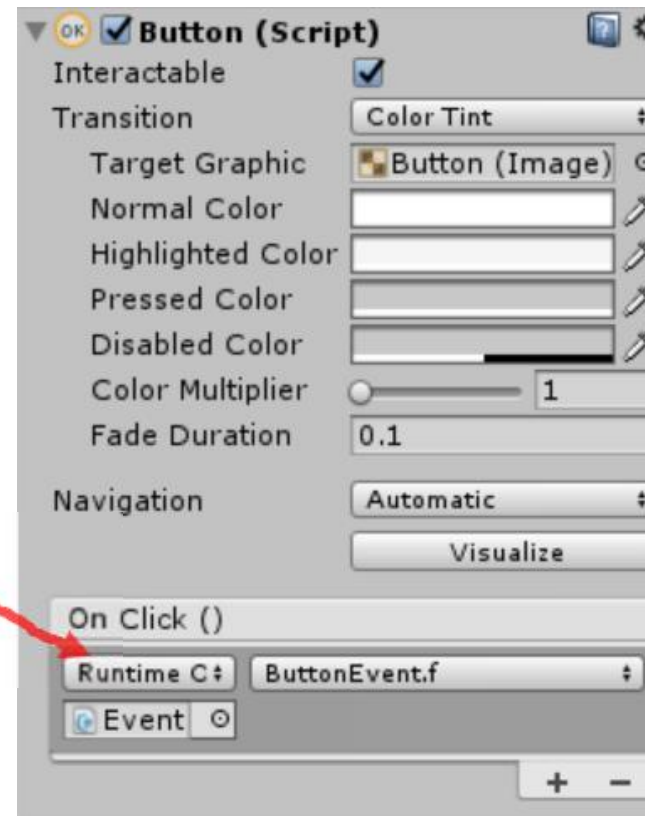
n A Button has an OnClick UnityEvent to define what it will do when clicked.



三、Interaction Components

① Button - Event

1) 面板中添加响应函数



三、Interaction Components

① Button - Event

2) 代码绑定响应函数

```
using UnityEngine.UI;  
using UnityEngine.Events;  
...  
...  
Button button;  
button.onClick.AddListener(new UnityAction(f));
```

三、Interaction Components

① Button - Event

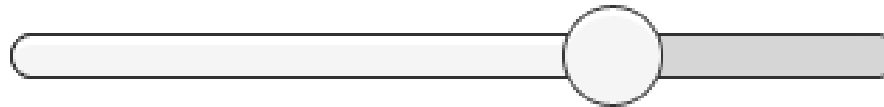
3) Lamda式绑定响应函数

```
Button button;  
button.onClick.AddListener(() =>  
{  
  
});
```


三、Interaction Components

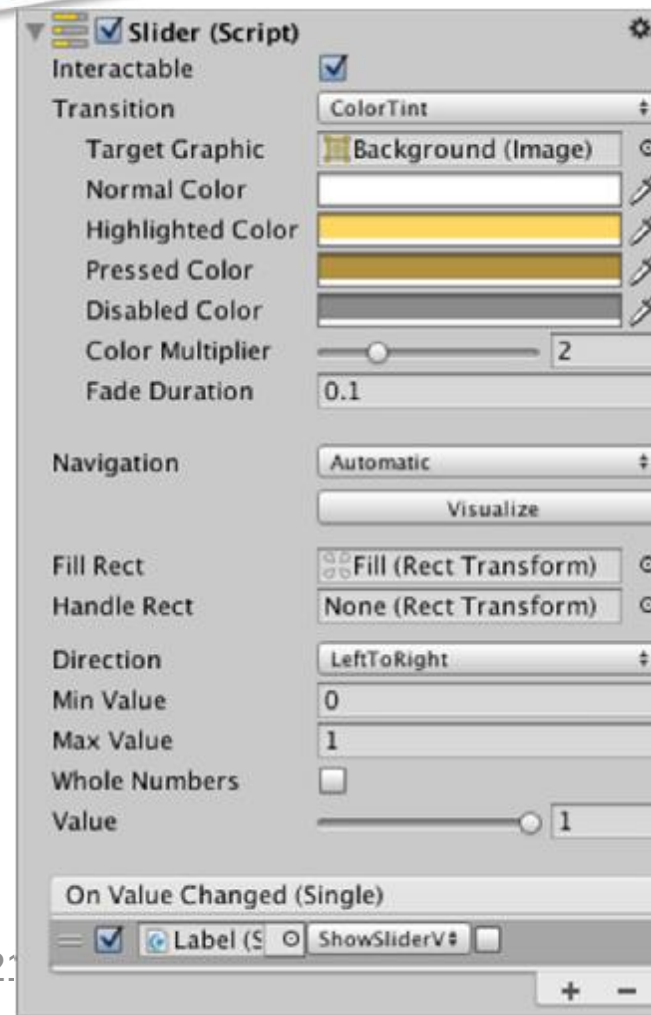
② Slider

- n A Slider has a decimal number **Value** that the user can drag between a minimum and maximum value. It can be either horizontal or vertical. It also has a **OnValueChanged** UnityEvent to define what it will do when the value is changed.



三、Interaction Components

② Slider



三、Interaction Components

② Slider

```
slider.onValueChanged.AddListener((float v) =>
{
    text.text = slider.value.ToString();
});
```

三、Interaction Components

③ Scrollbar

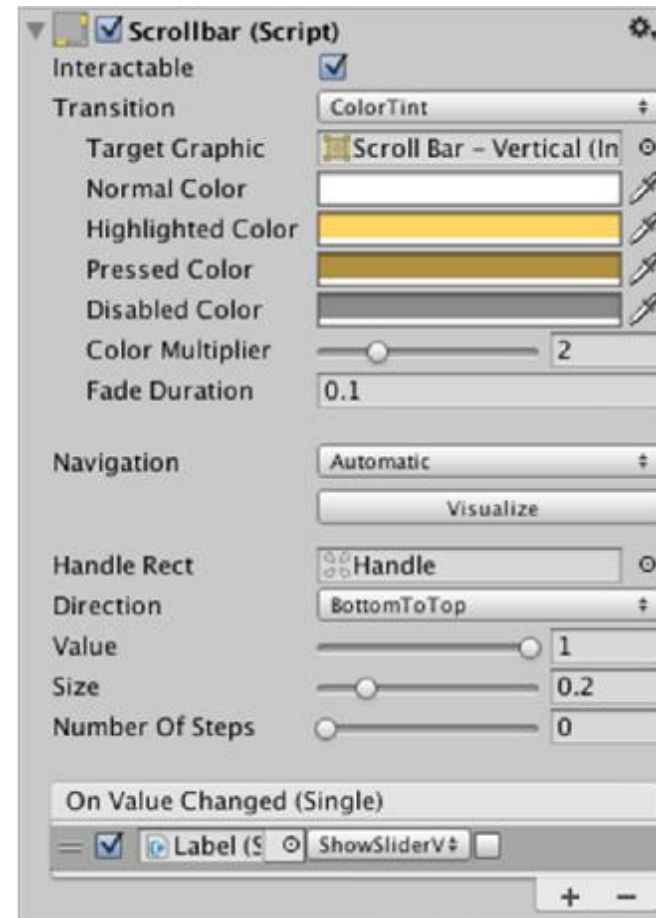
- n A Scrollbar has a decimal number **Value** between 0 and 1. When the user drags the scrollbar, the value changes accordingly.



三、Interaction Components

③ Scrollbar

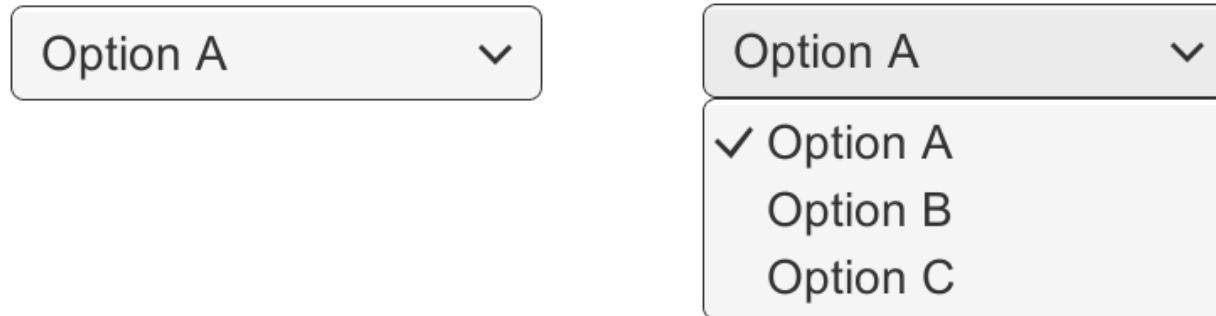
It also has a **OnValueChanged** UnityEvent to define what it will do when the value is changed.



三、Interaction Components

④ Dropdown

A Dropdown has a list of options to choose from. A text string and optionally an image can be specified for each option, and can be set either in the Inspector or dynamically from code. It has a **OnValueChanged** UnityEvent



三、Interaction Components

⑤ Input Field

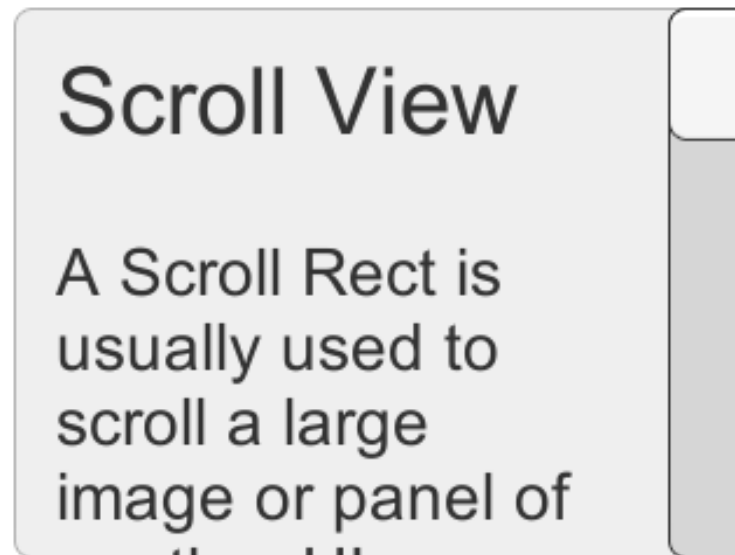
An Input Field is used to make the text of a [Text Element](#) editable by the user.

Enter text...

三、Interaction Components

⑥ Scroll View

A Scroll Rect can be used when content that takes up a lot of space needs to be displayed in a small area. The Scroll Rect provides functionality to scroll over this content.



四、 Animation Integration

- n Animation allows for each transition between control states to be fully animated using Unity's animation system.
- n This is the most powerful of the transition modes due to the number of properties that can be animated simultaneously.

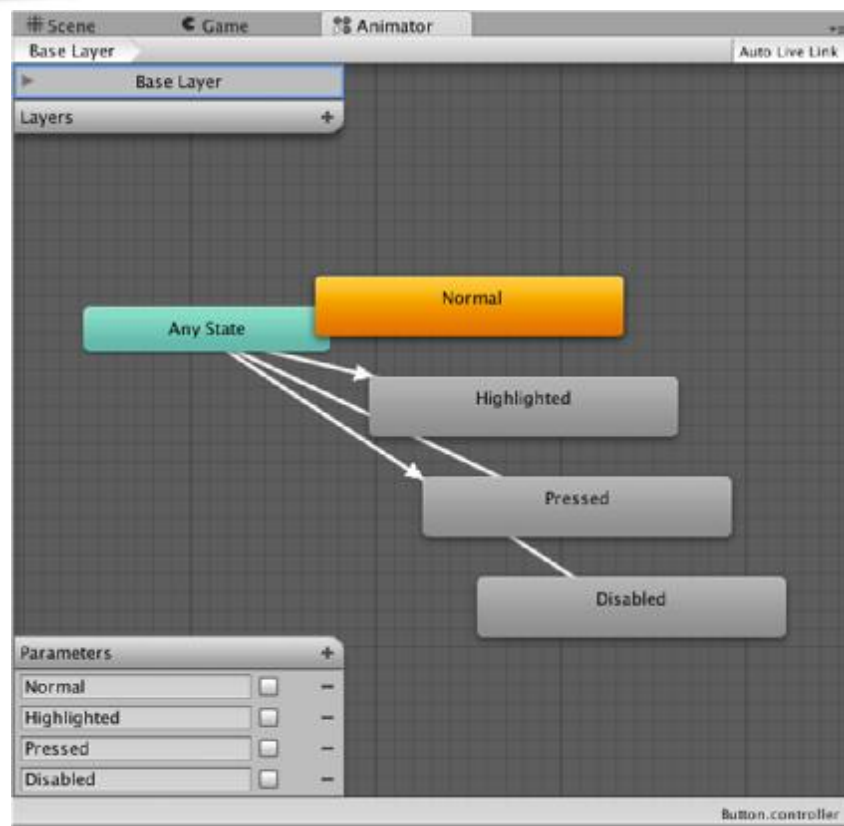
四、Animation Integration



四、Animation Integration

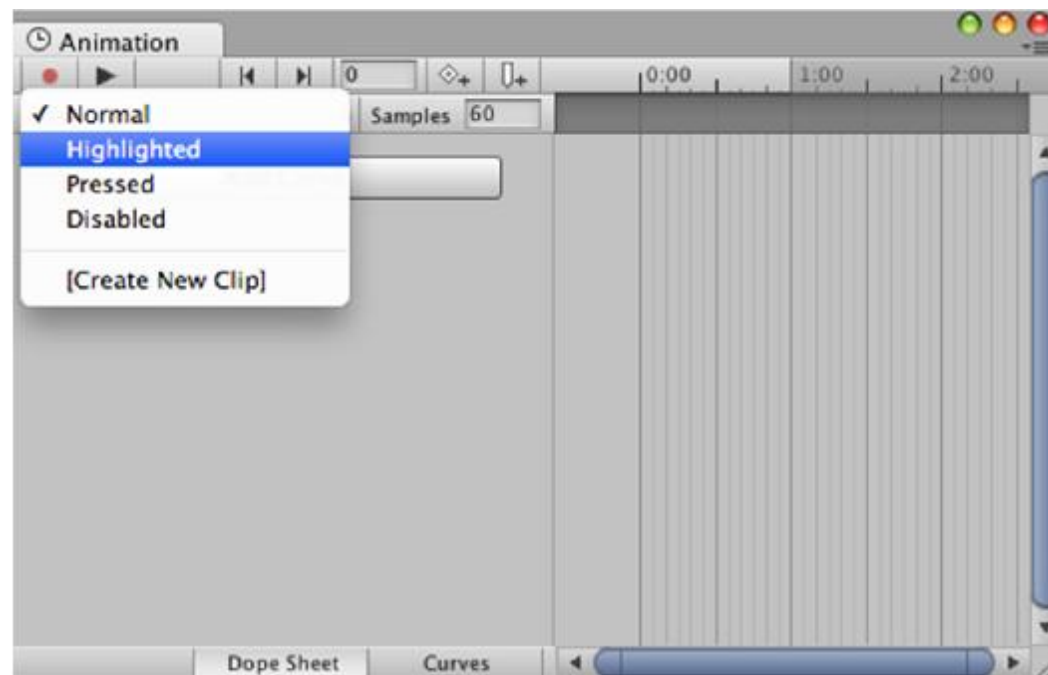
- n To use the Animation transition mode, an Animator Component needs to be attached to the controller element. This can be done automatically by clicking “Auto Generate Animation”. This also generates an Animator Controller with states already set up, which will need to be saved.
- n The new Animator controller is ready to use straight away. Unlike most Animator Controllers, this controller also stores the animations for the controller’s transitions and these can be customised, if desired.

四、Animation Integration



四、Animation Integration

- n For example, if a Button element with an Animator controller attached is selected, the animations for each of the button's states can be edited by opening the Animation window (**Window>Animation**).

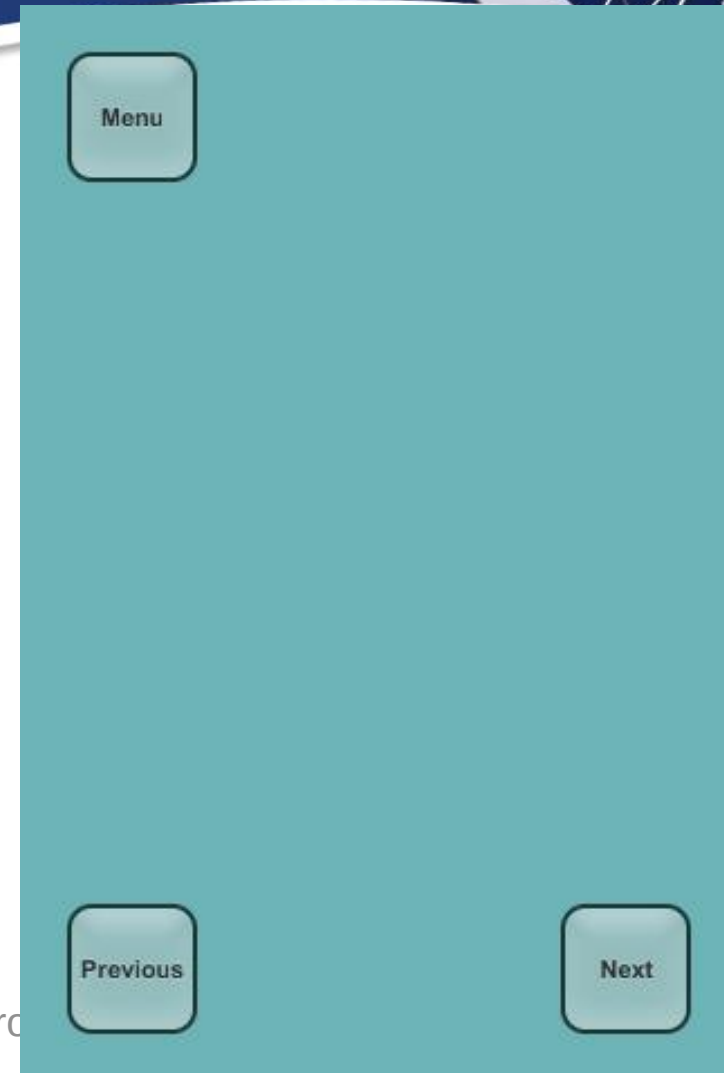


四、Animation Integration



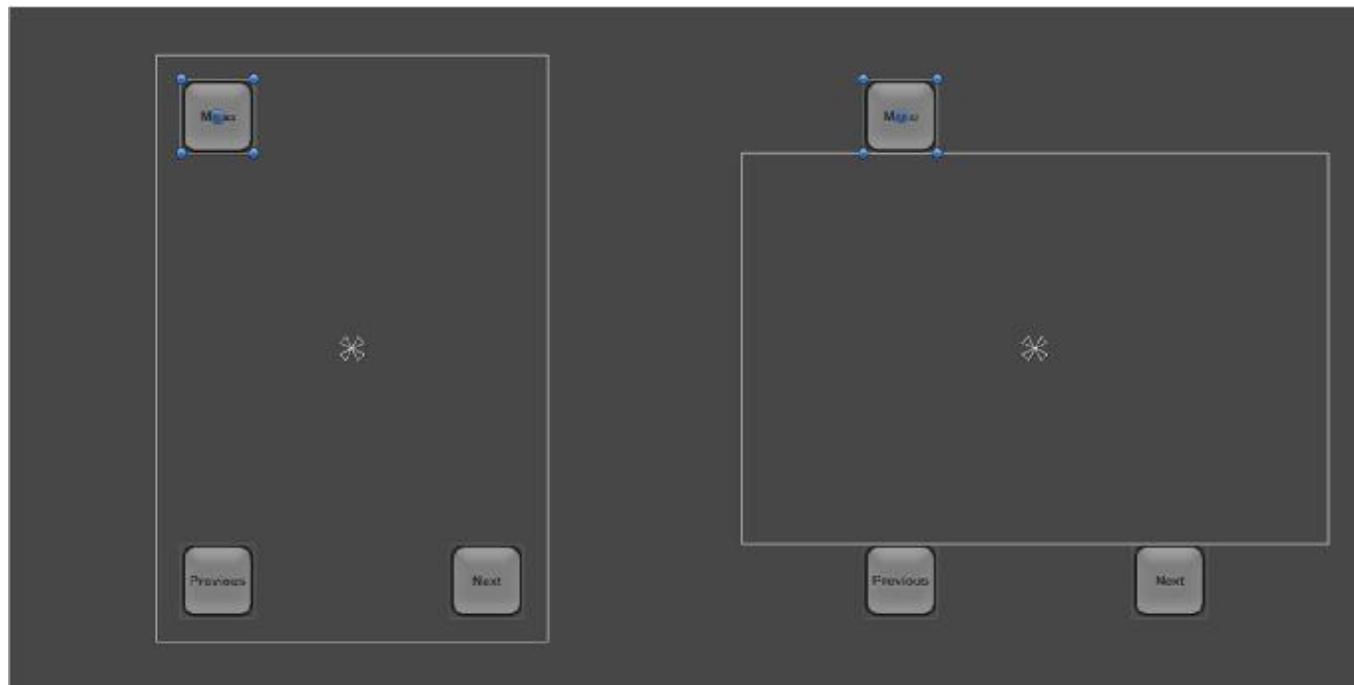
五、 Designing UI for Multiple Resolutions

- n Modern applications often need to support a wide variety of different screen resolutions and particularly UI layouts need to be able to adapt to that.



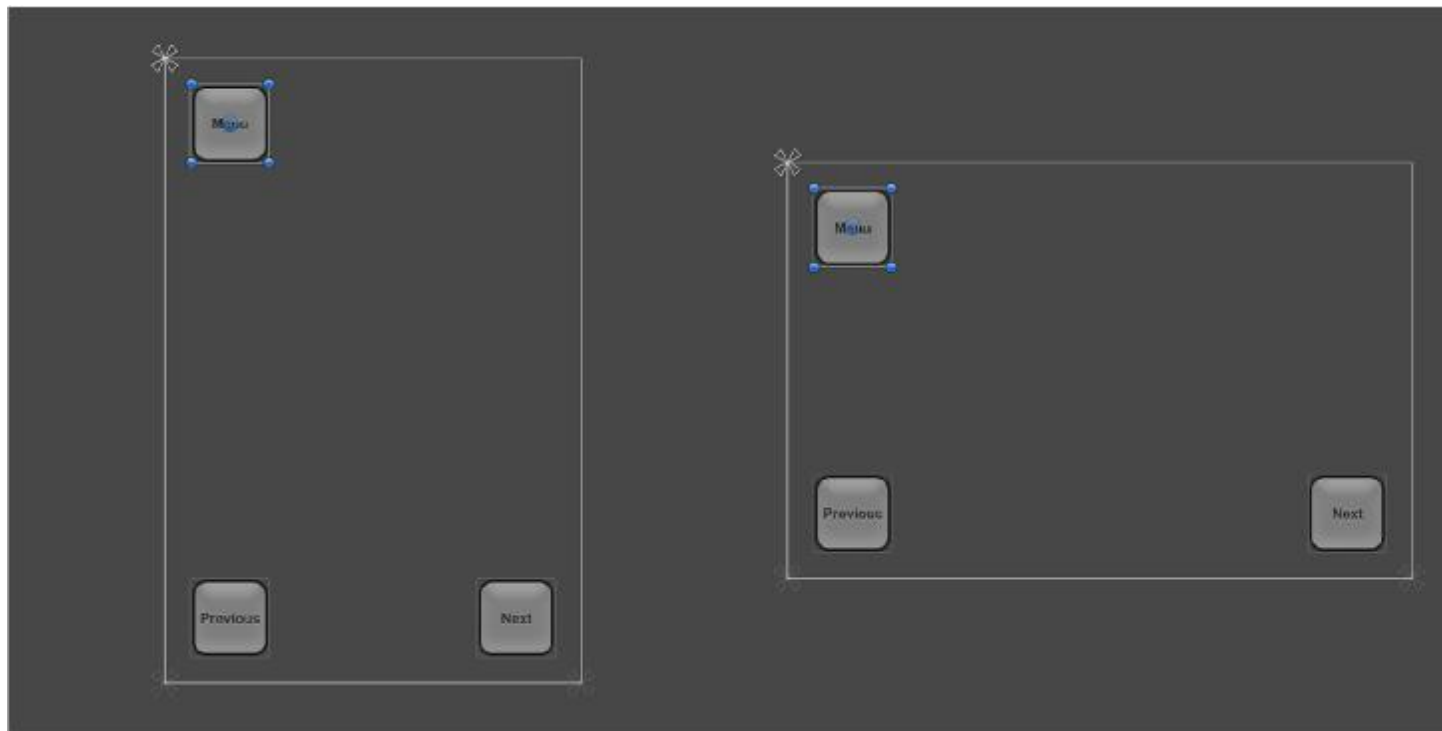
五、Designing UI for Multiple Resolutions

- UI elements are by default anchored to the **center** of the parent rectangle. If the resolution is changed to a landscape aspect ratio with this setup, the buttons may not even be inside the rectangle of the screen anymore.



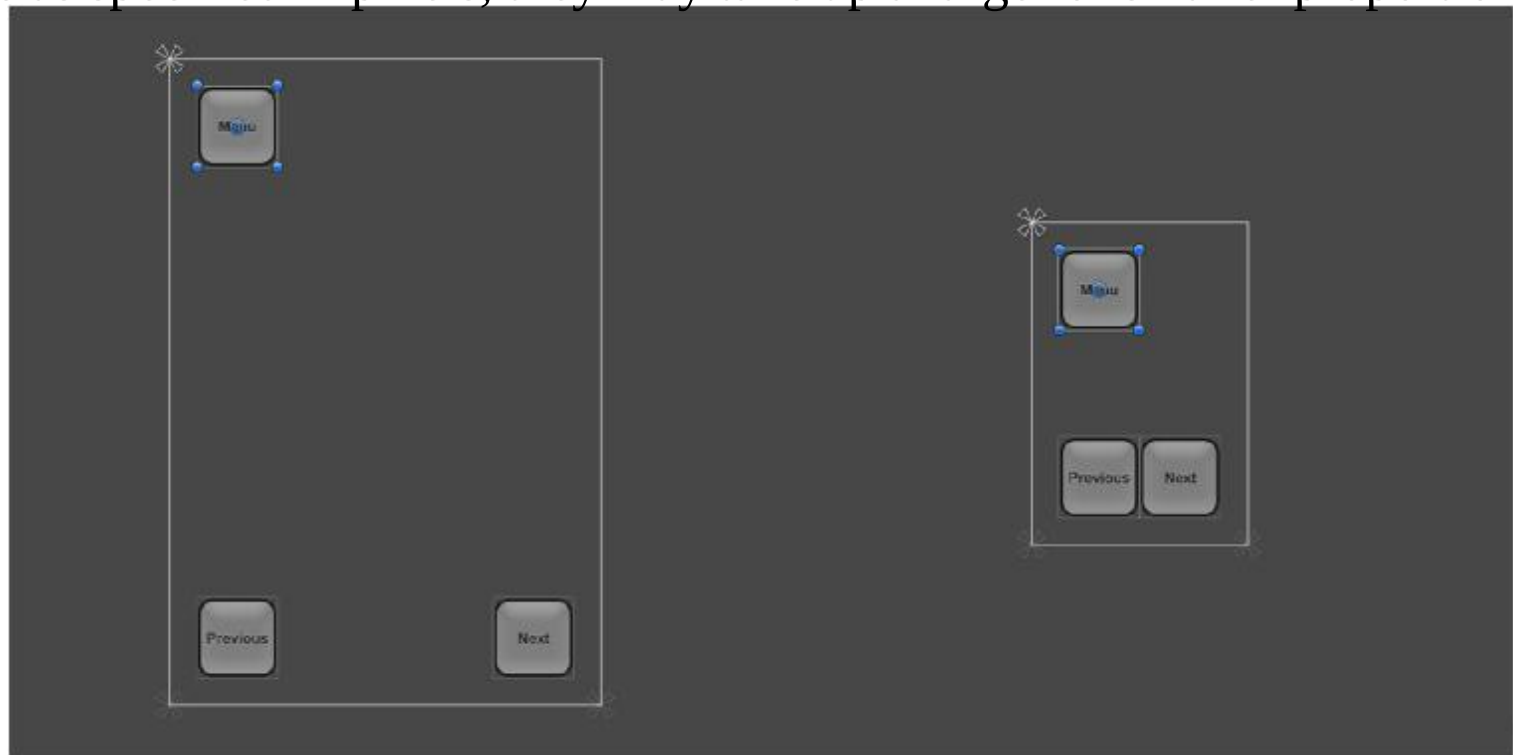
五、Designing UI for Multiple Resolutions

- Once the buttons have been anchored to their respective corners, they stick to them when changing the resolution to a different aspect ratio.



五、Designing UI for Multiple Resolutions

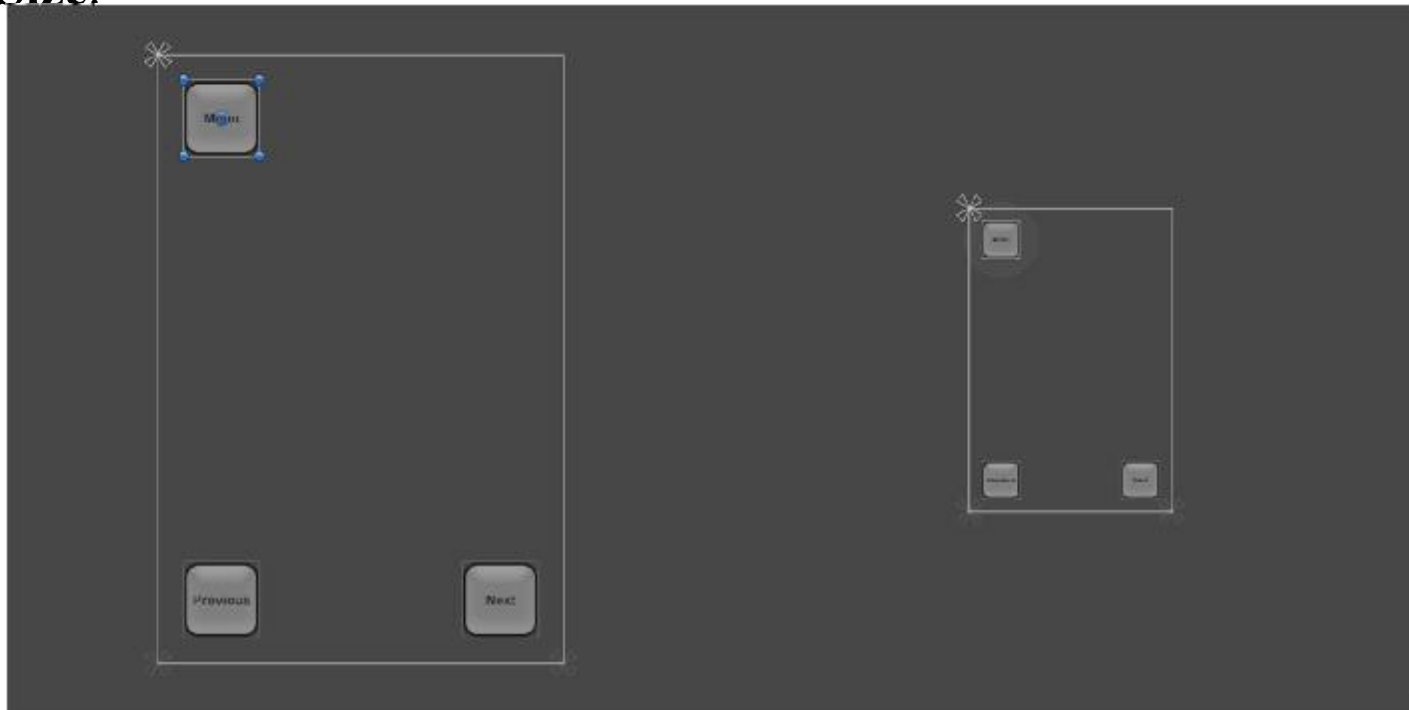
- When the screen size is changed to a larger or smaller resolution, the buttons will also remain anchored to their respective corners. However, since they keep their original size as specified in pixels, they may take up a larger or smaller proportion of the screen.



五、Designing UI for Multiple Resolutions

n Scaling with Screen Size

- n In the Canvas Scaler component, you can set its **UI Scale Mode** to **Scale With Screen Size**.

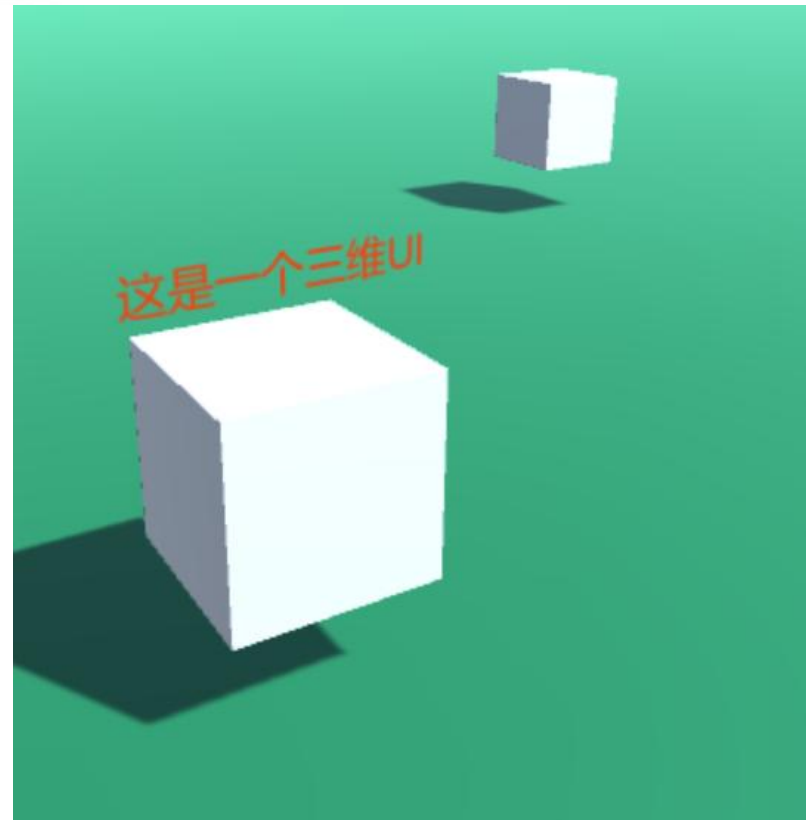


六、Creating a World Space UI

n Set the Canvas to World Space

n Select your **Canvas** and change the **Render Mode** to **World Space**.

六、Creating a World Space UI





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结束!